What count accuracy does and does not establish for own motion

8 October 2026 — source/static qualification; no joined law adopted

Scope. Source-supported CM and conditional numerical checks, followed by reviewer-derived finite-birth and objective limits. These are not a traffic simulation, reached metering comparison, full endogenous theorem or gradient certificate. The source audit distinguishes the supplied 2005 journal from the nearest cited 2004 thesis and unverified geometry manuscript.

1. The feedback quantity needs a definite operator

Use free speed $u > 0$, backward-wave magnitude $w > 0$, jam density $\kappa > 0$, and continuous density ρ:

$$F(\rho) = \min\{u\rho, w(\kappa - \rho)\}, \quad \rho_c = \frac{w\kappa}{u+w}, \quad V(\rho) = \begin{cases} u, & 0 \le \rho \le \rho_c, \\ w(\kappa/\rho - 1), & \rho_c < \rho \le \kappa. \end{cases} \tag{1}$$

Units: u, w, V in m/s; ρ, κ in equivalents/m; F in equivalents/s. On a fixed interior downstream support of length $\ell > 0$, compare

$$v_{\text{density}} = V\left(\frac{1}{\ell}\int_x^{x+\ell} \rho\, dy\right), \quad v_{\text{speed}} = \frac{1}{\ell}\int_x^{x+\ell} V(\rho)\, dy, \quad v_{\text{ratio}} = \frac{\int_x^{x+\ell} F(\rho)\, dy}{\int_x^{x+\ell} \rho\, dy}. \tag{2}$$

The last expression assumes positive mass and an instantaneous spatial diagnostic. It is not automatically the source's preceding-period Edie lane-choice estimate or a prescribed CM estimator.

For equal support halves, $u = 30$, $w = 5$, $\kappa = 3/20$, $\rho_1 = 1/100$ and $\rho_2 = 1/10$:

$$F_1 = 3/10, \quad F_2 = 1/4, \quad v_{\text{density}} = 95/11, \qquad v_{\text{speed}} = 65/4, \qquad v_{\text{ratio}} = 5. \tag{3}$$

Thus all three natural readings differ on the same own field. The example qualifies operator meaning; it is not hidden-information or control evidence.

The supplied 2006 Appendix A gives a constrained realized speed, desired-speed recurrence and motion of the form

$$v_j = \min\{v_j^{\text{dwn}}, v_j^{\text{des}}\}, \qquad v_{j+1}^{\text{des}} = v_j + a(v_j, \phi_j)\Delta t, \qquad \phi_{j+1} = \phi_j + v_j \Delta t. \tag{4}$$

The own downstream estimate is the missing specification, not an extra population. D04 supplies a density-average numerical precedent, not the complete adapted mandatory-node/source/encounter law.

2. A count bound alone does not close motion feedback

In a *source-free prescribed-path* diagnostic, $N_x = -\rho$. Compare two compatible count fields on a common support with

$$\|N_h - N\|_\infty \le \varepsilon_h, \qquad \bar{\rho} = \frac{N(t,x) - N(t,x+\ell)}{\ell}. \tag{5}$$

Endpoint subtraction gives

$$|\bar{\rho}_h - \bar{\rho}| \le \frac{2\varepsilon_h}{\ell}. \tag{6}$$

The triangular-FD speed function is globally Lipschitz on $[0,\kappa]$, with an admissible constant $L_V = w\kappa/\rho_c^2$: it is constant below ρ_c and its congested derivative has magnitude $w\kappa/\rho^2$. Consequently

$$|V(\bar{\rho}_h) - V(\bar{\rho})| \le \frac{2L_V\varepsilon_h}{\ell}. \tag{7}$$

J01's piecewise prescribed-path theorem bounds count error by $\varepsilon_h \le 3P\kappa\Delta x$ for its stated data class. For a mesh-dependent support $\ell = m\Delta x$, the resulting estimate is only

$$|V(\bar{\rho}_h) - V(\bar{\rho})| \le \frac{6P\kappa L_V}{m}. \tag{8}$$

This bound does not tend to zero with the mesh. It *does not prove* that actual speed errors fail to converge. A fixed positive physical support does make this particular bound vanish, but changes the support limit and cannot be silently interpreted as immediately downstream motion. Variable end supports, moving path errors and endogenous sources need additional work. Prescribed-path count convergence is not a joined feedback theorem.

3. Check a direct stationary boundary transcription

Let an existing boundary-born restriction have prescribed speed $v = u$, $\phi(t) = ut$, with empty/free-compatible receiving road, free downstream, and attempted incoming flow $0 < d \le Q = uw\kappa/(u+w)$. Ordinary compatible traffic has $F(\rho) \le u\rho$, so a zero-relative-passing condition at that moving path does not bind. Over the first interval the ordinary boundary can admit $d\Delta t$.

The specific transcription that holds the restriction at $x = 0$ with fixed boundary capacity zero for the entire interval instead admits zero:

$$\Delta C_{\text{moving}} = d\Delta t, \qquad \Delta C_{\text{held}} = 0, \qquad \Delta C_{\text{moving}} - \Delta C_{\text{held}} = d\Delta t > 0. \tag{9}$$

It is a finite-step count mismatch consistent with a vanishing mesh error, not a source-wide impossibility result. It exposes activation qualification for the candidate's mandatory-boundary transfer. No full-TST difference follows just from moving waiting mass outside versus inside before any exit occurs.

4. Finite population bounds births, not clearance

Use the admitted finite, forward-directed cell graph: through and discretionary moves advance one section; the finite ramp joins the mainline once. Assign a remaining-step rank r_a to each cell or external waiting location, with every accepted advance reducing rank by at least one and $0 \le r_a \le K$. No cyclic or backward population movement is admitted in this deduction.

For continuous masses m_a, define the nonnegative potential

$$P(t) = \sum_a r_a m_a(t). \tag{10}$$

Let $A_\infty < \infty$ be total offered mass after initialization and $H(T)$ the cumulative mass counted across all advancing moves through time T. Conservation and decreasing ranks imply

$$H(T) \le P(0) + KA(T) \le P(0) + KA_\infty. \tag{11}$$

The constants K and S refer to the declared finite grid. This is not a uniform record-count bound under arbitrary mesh refinement. Moving waves do not move population backward. A flow split is debited once per actual advance, in accordance with the source IT update; it is not counted as both a same-cell lateral move and a second longitudinal move.

Let S be the finite number of descriptor-generating streams, threshold $\delta > 0$, initial progress $r_s^0 \in [0, \delta)$, and accepted generating transfer $\eta_s(T)$. E1's cumulative floor gives

$$B_s(T) = \left\lfloor \frac{r_s^0 + \eta_s(T)}{\delta} \right\rfloor, \qquad \sum_s \eta_s(T) \le H(T). \tag{12}$$

Hence, including all initial remainders and retaining all distinct births,

$$B_{\text{new}}(T) = \sum_s B_s(T) \le \frac{P(0) + KA_\infty}{\delta} + S. \tag{13}$$

No threshold integerizes traffic mass. The bound is a finite-geometry feasibility deduction, not a numerical cost or record-budget requirement. It does not prove that traffic or restrictions eventually leave; a stationary blocking update could still fail to clear despite finitely many births.

5. Finite-time performance and derivatives are separate claims

Under the same offered arrivals and initial population, conservative full-system inventory includes outside waiting, ramp and mainline:

$$M(t) = M_0 + A(t) - E(t), \qquad J_T = \int_0^T M(t)\,dt. \tag{14}$$

E counts all exits. If a conditional comparison establishes a uniform exit count error $|E_h - E| \le \varepsilon$ on $[0, T]$, then

$$|J_{T,h} - J_T| \le T\varepsilon. \tag{15}$$

This says nothing about the missing tail beyond T. Full J requires clearance or a qualified integrable tail. Normalizing by common total offered and initial mass rescales the bound but does not remove that obligation.

Even uniform objective convergence does not guarantee its gradients. For an arbitrary smooth f and $h > 0$, a generic calculus example is

$$f_h(\theta) = f(\theta) + h\sin(\theta/h), \quad |f_h - f| \le h, \quad f_h'(\theta) = f'(\theta) + \cos(\theta/h). \tag{16}$$

This is not a claimed traffic formula. It demonstrates the logical limit of promoting count/objective accuracy into derivative accuracy. Threshold births and active-set changes introduce additional obligations in the actual task.

Disposition. The physical source family and a concrete numerical feedback precedent are supported. The displayed source-derived candidate is not frozen: activation, feedback support/phase, joined accepted sources and simultaneous paths require a declared completion. Finite births, finite-time count/TST checks and a causal interface do not establish full clearance, nontrivial metering leverage, record necessity or useful gradients.